Fourier Analysis: M. Math II: Mid Semestral Examination

September 10, 2025.

Maximum Marks 40 | Duration 2 hrs.

State any result you use. Brief and clearly expressed answers will be appreciated.

(1) Take a function $f \in C^1(\mathbb{T})$. Show that the Fourier series of f is absolutely convergent, without using the result about Lipschitz functions. [5]

(2) Find the complete range of p, q such that for all $f \in C(\mathbb{T})$,

$$\|\widehat{f}\|_{\ell^q} \leq C\|f\|_{L^p(\mathbb{T})}.$$

$\hat{f}(\xi) = \int f(x)\, e^{-ix\xi}$

[5]

(3) Suppose for $f \in L^1(\mathbb{T})$, $\widehat{f}(n) \neq 0$ for all $n \in \mathbb{Z}$. Let

$$V(f) = \text{Span}\{\tau_\alpha f \mid \alpha \in [0, 2\pi]\},$$

where $\tau_\alpha f(\theta) = f(\theta - \alpha)$. Show that $V(f)$ is dense in $L^1(\mathbb{T})$. [Hint: What is the dual of $L^1(\mathbb{T})$?] [5]

(4) (a) Let t be a trigonometric polynomial and $f \in L^1(\mathbb{T})$. Show that $f * t$ is a trigonometric polynomial.

(b) Take $f \in L^1(\mathbb{T})$ such that f is not a.e. identical with a C^1-function. Show that $\widehat{f}(n) \neq 0$ for infinitely many $n \in \mathbb{Z}$.

[3+2]

(5) Show that $\sup_N \|S_N f\|_{L^1 - L^1} = \infty$. [5]

P.T.O.

(6) (a) State Hausdorff-Young inequality on $\mathbb{R}^n$.

(b) Show how you can use this to define Fourier transform of a function $f \in L^p(\mathbb{R})$ where $1 < p < 2$.

(c) Recall that Fourier transform of this $f \in L^p(\mathbb{R})$ can also be defined by writing $f = f_1 + f_2$ where $f_1 \in L^1$ and $f_2 \in L^2$. Show that Fourier transforms of f (denote them by $\mathcal{F}_1, \mathcal{F}_2$) defined in these two ways are same as measurable functions. [2+3+5 = 10]

(7) Consider the function $f(x) = x^2, x \in \mathbb{R}$. Find its Fourier transform. [5]

(8) Let T is a self-adjoint linear operator which is $p - p$. Show that it is $2 - 2$.

[5]

(9) Suppose $f \in L^1(\mathbb{R})$ satisfies $f(x) = 0$ for $x < 0$. Find the domain in the complex plane where $\widehat{f}$ can be extended as a holomorphic function. [5]